

Lightweight AI for UAV-Mounted RIS: An Overview

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Abstract—Unmanned Aerial Vehicles (UAV)-mounted Reconfigurable Intelligent Surfaces (RIS) have emerged as a promising architecture for enhancing wireless coverage, spectral efficiency, and energy performance in 6G networks. By combining programmable electromagnetic wave manipulation with aerial mobility, UAV-RIS systems enable dynamic blockage mitigation, adaptive beamforming, and flexible deployment across terrestrial, maritime, and satellite-integrated environments. However, joint optimization of UAV trajectory, RIS phase configuration, and resource allocation incurs high computational complexity, which is incompatible with the strict energy and onboard processing constraints of UAV platforms. Lightweight AI techniques offer practical solutions to this challenge. Hence, this paper provides a comprehensive overview of lightweight AI techniques for UAV-mounted RIS systems, including Reinforcement Learning (RL), meta-learning, Federated Learning (FL), Multi-Armed Bandits (MAB), and energy-aware optimization. We present a detailed taxonomy and comparative analysis of existing work, highlight computational-energy trade-offs, and identify open research challenges for scalable, energy-efficient airborne intelligent surfaces. Furthermore, we present a case study demonstrating the effect of MAB schemes on throughput and energy efficiency in UAV-mounted RIS.

Index Terms—UAV mounted RIS; Lightweight AI; RL, MAB, Meta learning;

I. INTRODUCTION

Reconfigurable Intelligent Surfaces (RIS) have been recognized as a key enabler of programmable wireless environments for beyond-5G and 6G systems [1], [2]. RIS enables controllable reflection of electromagnetic waves, improving coverage and Energy Efficiency (EE) without requiring active Radio Frequency (RF) chains. While early research focused on static, ground-mounted RIS, recent developments have introduced UAV-mounted RIS systems, where programmable reflecting surfaces are integrated into aerial platforms [3]–[7].

UAV-mounted RIS systems provide unique advantages over terrestrial RIS installations. By leveraging 3D mobility, UAV-RIS can dynamically adjust both horizontal position and altitude to optimize channel conditions [8]–[11]. This flexibility enables rapid deployment in emergency scenarios [4], extended

connectivity in rural or maritime environments [10], and integration with satellite and Non-Terrestrial Networks (NTN) [12]–[16]. Also, UAV-RIS has been explored for Integrated Sensing and Communication (ISAC) [17], secure communication [18], and Mobile Edge Computing (MEC) [19], [20].

Moreover, recent advancements in RIS hardware architectures have introduced multiple RIS types with distinct operational characteristics, as highlighted in Table I. Figure.1 presents the four main types of aerial RIS used in different scenarios. Conventional passive RIS relies solely on controllable Phase Shifts (PSs) without signal amplification, offering low power consumption and hardware simplicity [2]. Furthermore, active RISs incorporate low-power amplifiers within reflecting elements to enhance signal strength, albeit at the cost of increased energy consumption and hardware complexity [11], [18]. Simultaneously Transmitting and Reflecting RIS (STAR-RIS) architectures enable simultaneous signal transmission and reflection to serve users on both sides of the RIS panel [6], [12]. This significantly improves spatial coverage and flexibility in aerial RIS-assisted deployments. Furthermore, hybrid RIS architectures that combine passive, active, and STAR components aim to balance EE and performance enhancement [21], [22]. Emerging Beyond-Diagonal RIS (BD-RIS) and coupled-element designs introduce non-diagonal scattering matrices that enable more flexible wave manipulation and spatial multiplexing, further expanding the design space of airborne RIS systems [15], [16]. These diverse RIS architectures directly affect the computational and energy requirements of UAV-mounted RIS optimization, underscoring the importance of lightweight AI techniques for real-time control.

Nevertheless, UAV-mounted RIS systems introduce several technical challenges. Mobility and altitude variation lead to time-varying cascaded BS-RIS-user channels and severe Doppler effects [7], [9]. Accurate and frequent Channel State Information (CSI) acquisition becomes costly, particularly when the RIS dimension is large. Moreover, the joint optimization of UAV trajectory, altitude, RIS phase configuration, transmission scheduling, and, in some cases, power allocation or amplification (for active/STAR-RIS) results in a tightly coupled, non-convex, high-dimensional optimization problem

TABLE I
COMPARISON OF RIS ARCHITECTURES AND THEIR LIGHTWEIGHT AI REQUIREMENTS FOR UAV-MOUNTED SYSTEMS

RIS Type	Operation Mode	Power & Hardware Cost	Beamforming Capability	Optimization Complexity	Typical AI Approach
Passive RIS [2]	Reflection only	Low	Moderate (double fading effect)	Moderate	RL / MAB
Active RIS [11], [18]	Amplified reflection	High	High gain (attenuation mitigation)	High (power + phase control)	Energy-aware DRL
STAR-RIS [6], [12]	Simultaneous transmit & reflect	Medium-High	Dual-side coverage flexibility	High (dual-mode control)	Meta-RL / DRL
Hybrid RIS [21], [22]	Passive + active elements	Medium	Gain-performance trade-off	High (mode selection + amplification)	Hybrid RL + model compression
BD-RIS [15], [16]	Beyond-diagonal coupled scattering	High	Enhanced spatial multiplexing	Very High (matrix optimization)	Deep unfolding / advanced RL

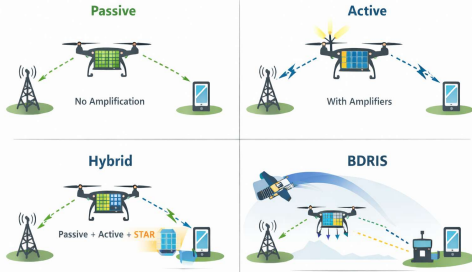


Fig. 1. UAV-mounted RIS architectures

[23], [24]. The complexity further increases when advanced RIS architectures such as hybrid or STAR-RIS are employed, as additional transmission/reflection modes must be optimized simultaneously. From a practical perspective, UAV platforms are inherently constrained by limited battery capacity, strict payload weight limits, and restricted onboard processing capability [4], [25]. UAV-mounted RIS controllers cannot support iterative convex solvers or large-scale Deep Neural Networks (DNNs) that require extensive floating-point operations. Conventional optimization techniques, including Alternate Optimization (AO) and Successive Convex Approximation (SCA), often involve iterative matrix operations whose complexity scales with the number of RIS elements, making them unsuitable for real-time aerial deployment. As RIS size grows to support higher spatial resolution, computational overhead increases dramatically, exacerbating energy consumption and latency. These limitations motivate the adoption of lightweight AI techniques, rather than generic Deep Learning (DL) frameworks as a necessity for enabling real-time inference under stringent energy and hardware constraints. Techniques such as Reinforcement Learning (RL) and deep RL (DRL) enable model-free decision-making without explicit channel modeling [23], [24], while meta-learning facilitates rapid adaptation to dynamic channel conditions at reduced retraining cost [26]. Federated learning (FL) allows distributed optimization across multiple UAV-RIS nodes without excessive communication overhead [27], and Multi-Armed Bandit (MAB) frameworks provide low-complexity online decision-making suitable for mmWave and dynamic spectrum scenarios [28]. This motivates a systematic investigation of lightweight AI methodologies tailored specifi-

cally for UAV-mounted RIS architectures.

Hence, this paper surveys lightweight AI techniques for UAV-mounted RIS systems and provides comparative insights into their performance and complexity characteristics. Also, it presents a case study that highlights the importance of these techniques. The remainder of this paper is organized as follows. Section II explores different lightweight AI techniques used in UAV-mounted RIS networks. Section III lists important application scenarios for lightweight AI in aerial RIS networks. Section IV discusses open research challenges and future directions. Section V introduces a case study that shows the effect of MAB on the performance of UAV-mounted RIS scenario. Finally, Section VI concludes the paper.

II. SUMMARY OF LIGHTWEIGHT AI TECHNIQUES FOR UAV-MOUNTED RIS SYSTEMS

The principal lightweight AI methodologies applicable to UAV-mounted RIS systems are divided into four main categories as follows:-

A. Model Efficiency techniques

1) *Model Compression: Pruning and Quantization:* Model Compression significantly reduces NN complexity while preserving performance. Pruning eliminates redundant weights or filters, reducing memory access and inference latency. Quantization reduces numerical precision, thereby lowering computational cost and energy consumption. In UAV-mounted RIS systems, compressed DRL or BeamForming (BF) models can operate efficiently on embedded processors. These techniques are especially important when controlling large-scale RIS panels, where inference complexity scales with the number of reflecting elements. Model compression enhances onboard feasibility without sacrificing adaptability.

2) *Knowledge Distillation:* It transfers knowledge from a large, computationally intensive "teacher" model to a smaller "student" network suitable for resource-constrained deployment. In UAV-mounted RIS systems, most existing work relies on DRL or complex joint-optimization frameworks [5], [23], [24]. While these approaches achieve strong performance, their computational overhead may limit real-time onboard implementation. Large offline-trained DRL models developed at ground stations could be distilled into compact student models for embedded aerial controllers. This would enable near-optimal

TABLE II
COMPARISON OF LIGHTWEIGHT AI TECHNIQUES FOR UAV-MOUNTED RIS SYSTEMS

Category	Technique	Computational Complexity	Energy Efficiency	Adaptability	Onboard Feasibility
Model Efficiency	Pruning	Low	High	Medium	High
Model Efficiency	Quantization	Very Low	Very High	Medium	Very High
Model Efficiency	Knowledge Distillation	Low	High	Medium-High	High
Model Efficiency	TinyML	Very Low	Very High	Medium	Very High
Adaptive Decision-Making	Lightweight RL/DRL	Medium	Medium	High	High
Adaptive Decision-Making	Meta-Learning	Medium	Medium	Very High	Medium
Adaptive Decision-Making	Transfer Learning	Medium	Medium	Very High	Medium
Adaptive Decision-Making	MAB	Low	High	Medium	Very High
Adaptive Decision-Making	Edge Computing	Medium	High	High	High
Topology-Aware Distributed	Federated Learning	Medium	Medium	High	Medium
Topology-Aware Distributed	Sparse GNN	Medium	Medium	High	Medium
Hybrid Model-Driven	Deep Unfolding	Low-Medium	High	High	High

trajectory and phase configuration control with significantly reduced inference complexity.

3) *Tiny Machine Learning (TinyML) and Onboard Edge Intelligence*: TinyML enhances aerial RIS intelligence, enabling lightweight phase prediction, channel classification, and simple trajectory adaptation. It focuses on deploying ML models on ultra-low-power micro-controllers with minimal memory and processing requirements [29]. In UAV-mounted RIS systems, most existing AI-driven optimization frameworks rely on DRL or centralized training approaches [5], [23], [24], which achieve strong performance in simulation environments but may be practically computationally intensive. By eliminating reliance on centralized processing, TinyML reduces communication overhead and latency. This is beneficial in emergency deployments or NTN-integrated scenarios where backhaul connectivity may be limited. Given the strict energy and payload constraints of UAV platforms [4], [25], TinyML offers a practical pathway toward fully autonomous UAV-mounted RIS systems.

B. Adaptive Decision making methods

1) *RL and DRL*: RL has become a dominant framework for UAV-mounted RIS optimization due to its ability to handle model-free and dynamic control problems. RL-based methods jointly optimize UAV trajectories and RIS phase configurations via environmental interactions [23], [30], [31]. DRL extends this capability using neural function approximators to scale to larger state spaces [5], [17], [24], [25], [32]. To enable onboard deployment, lightweight DRL variants reduce network depth, discretize action spaces, and compress state representations. Energy-aware reward formulations further integrate battery constraints into learning policies [25], [33].

2) *Meta-Learning and Transfer Learning*: Meta-learning enhances generalization by enabling rapid adaptation to new environments with minimal retraining. Given the dynamic propagation conditions of UAV-mounted RIS systems, Meta-RL frameworks are particularly effective [26]. Transfer learning techniques reuse pretrained DRL policies in new deployment scenarios, significantly reducing computational overhead and training time [22]. These approaches are valuable in heterogeneous environments such as maritime communications, NTN-integrated systems, and urban mobility scenarios.

3) *Multi-Armed Bandit (MAB) Methods*: MAB algorithms provide a computationally efficient stateless RL framework. It

enables fast decision-making with minimal overhead. In UAV-mounted RIS systems, MAB has been applied to mmWave beam selection, RIS element allocation, and WiGig enhancement [28], [34], [35]. Energy-aware MAB schemes further incorporate battery constraints [36], [37]. MAB techniques are well-suited to real-time onboard implementation.

C. Topology-aware distributed methods

1) *Sparse Graph Neural Networks (GNNs)*: GNNs effectively model interactions among UAVs, users, and RIS elements. However, full GNN implementations may be computationally expensive. Sparse GNN architectures reduce message-passing operations by limiting neighborhood connectivity. In multi-UAV RIS deployments [38], sparse GNNs enable scalable topology-aware resource optimization while maintaining manageable computational cost. These approaches are suitable for both cooperative and distributed aerial networks.

2) *Federated Learning*: FL supports decentralized model training across multiple UAV nodes without sharing raw data [27]. This reduces communication overhead and enhances privacy. In cooperative UAV-mounted RIS systems [38], FL enables distributed optimization of trajectories and phases while reducing the signaling overhead. FL is particularly effective in large-scale aerial swarms or NTN-integrated architectures.

3) *Edge Computing*: Instead of performing full-scale trajectory and phase optimization onboard the UAV, edge-assisted architectures allow real-time coordination between aerial RIS platforms and ground-based MEC units [19], [20]. This reduces onboard processing burden, lowers energy consumption, and improves scalability. Recent work has further investigated EE UAV-mounted RIS-assisted MEC systems, in which joint optimization of communication, computation, and reflection control is performed to maximize overall system efficiency [8]. By combining edge computing with lightweight AI techniques, UAV-mounted RIS networks can achieve low-latency decision-making while preserving battery life and maintaining high adaptability in dynamic 6G environments.

D. Hybrid model-driven learning

1) *Deep Unfolding*: It integrates iterative optimization algorithms into the layers of structured NNs. Each layer corresponds to an iteration of a classical optimization algorithm, preserving

interpretability while reducing parameter count [39]. For UAV-mounted RIS phase optimization, deep unfolding provides a balance between model-based precision and data-driven flexibility. Compared with generic DNNs, unfolding architectures have lower computational complexity and faster convergence, making them suitable for embedded UAV controllers.

III. APPLICATION DOMAINS OF LIGHTWEIGHT AI IN UAV-MOUNTED RIS SYSTEMS

This section discusses the major application domains in which lightweight AI techniques are employed to optimize UAV trajectory, RIS phase configuration, energy management, and distributed coordination. Table III summarizes recent application domains and utilized AI solutions.

1) **Simultaneous Wireless Information and Power Transfer (SWIPT) and Energy Harvesting (EH)**: SWIPT, when combined with UAV-mounted RIS, enables joint data transmission and EH. Active and multifunctional RIS architectures are particularly beneficial in such scenarios, as they enhance both the reflected-signal strength and the harvested EE [18], [41]. Learning-based RIS element allocation and SWIPT optimization strategies have been proposed to balance throughput and harvested energy [41]. Security-aware SWIPT designs further enhance system resilience while maintaining EE [18]. Furthermore, the multi-objective nature of SWIPT systems, which simultaneously optimize data rate, harvested power, and UAV battery sustainability, results in high-dimensional, non-convex optimization problems. Therefore, Lightweight AI techniques are essential to enable real-time control under strict energy constraints while maintaining balanced system performance.

2) **Mobile Edge Computing (MEC)**: MEC enhances UAV-mounted RIS systems by enabling computational offloading to nearby edge servers. Instead of executing all optimization and learning tasks on board, the UAV computing platform collaborates with MEC servers, thereby offloading computationally intensive tasks to edge infrastructure to reduce latency and processing overhead [19], [20]. More recent frameworks jointly integrate UAV trajectory planning, computation task scheduling, and reflection control, in which the phase configuration of RIS elements is dynamically optimized to manipulate the wireless propagation environment within energy-efficient UAV-mounted RIS-assisted MEC systems. Thus, MEC acts as a system-level enabler that supports lightweight AI and optimization algorithms in computationally demanding scenarios.

3) **Satellite and NTN Integration**: The integration of UAV-mounted RIS with satellite systems supports NTN architectures and Space–Air–Ground Integrated Networks (SAGIN). In such configurations, UAV-RIS platforms assist LEO satellite communication by improving beam alignment and mitigating propagation loss [12]–[14], [42]. Joint optimization across satellite beams, UAV mobility, and RIS phase control yields highly coupled, computationally intensive problems. The presence of Doppler shifts and large-scale MIMO beamforming further increases complexity. Lightweight AI techniques are crucial to ensure real-time feasibility and scalable coordination between satellite and aerial layers while maintaining manageable on-board complexity.

4) **Maritime Communications**: It is characterized by sparse infrastructure, long propagation distances, and vulnerability to interference and jamming. UAV-mounted RIS systems enhance signal coverage by dynamically adjusting aerial positioning and reflective BF [10]. However, the dynamic sea environment introduces rapidly varying channel conditions and Doppler effects. Lightweight RL and energy-aware optimization frameworks have been proposed to jointly optimize UAV trajectory and RIS PSs for secure maritime sensing and communication [11]. Anti-jamming and EH-based DRL strategies further enhance robustness and sustainability in maritime deployments [33]. Hence, lightweight AI is essential to enable adaptive BF and secure control while preserving UAV battery life during long-duration missions.

5) **ISAC**: It introduces additional complexity because sensing objectives (e.g., detection accuracy and radar resolution) must be balanced with communication metrics (e.g., throughput and latency). This multi-objective optimization significantly increases computational requirements. Lightweight AI techniques are critical for enabling real-time ISAC control while preserving EE and onboard feasibility. UAV-mounted RIS platforms are particularly suitable for ISAC due to their aerial mobility and programmable electromagnetic response. By intelligently adjusting the UAV’s trajectory and the RIS phase configuration, UAV-RIS systems can simultaneously enhance communication throughput and sensing accuracy. Recent studies have investigated DRL-based EE UAV-RIS-assisted ISAC systems, where BF, sensing accuracy, and UAV trajectory are jointly optimized under energy constraints [17]. Hybrid-RIS-enabled UAV-assisted ISAC frameworks leveraging transfer-learning-based DRL have also been proposed to improve adaptability across dynamic environments [22].

IV. CHALLENGES AND FUTURE DIRECTIONS

Despite rapid progress in UAV-mounted RIS systems and AI-driven optimization, several fundamental challenges remain open. In particular, integrating lightweight AI with aerial mobility, programmable reflection¹ and energy-constrained platforms poses unique challenges that warrant further investigation.

1) **Scalability with Large-Scale RIS Arrays**: As the number of RIS elements increases, the dimensionality of the optimization problem grows significantly. Jointly with communication metrics (e.g., throughput and latency) and scheduling, it quickly becomes computationally prohibitive. Most existing work relies on DRL, which scales poorly as state and action dimensions increase. Hence, lightweight AI solutions such as pruning, quantization, and model-driven learning have not yet been systematically integrated into large-scale RIS optimization. Developing scalable architectures that maintain performance while controlling computational complexity remains a critical research challenge.

2) **Real-Time Adaptation under High Mobility**: UAV-mounted RIS systems operate in highly dynamic environments

¹The phase shifts of RIS elements are dynamically adjusted to control signal propagation

TABLE III
SUMMARY OF AI-BASED TECHNIQUES FOR UAV-MOUNTED RIS OPTIMIZATION

Ref.	Application Scenario	AI Technique	Optimization Target	RIS Type	Energy-Aware
[5]	mmWave-NOMA	DRL	Trajectory + Phase Control	Passive	No
[17]	ISAC Systems	DRL	ISAC Optimization	Hybrid	Yes
[25]	IoT Networks	DRL	Energy-Efficient Phase Control	Passive	Yes
[23]	General UAV-RIS	RL	Trajectory + Phase Shift	Passive	No
[32]	AoI Optimization	DRL	AoI-Aware STAR-RIS Control	STAR-RIS	No
[24]	CoMP-NOMA	DRL	Trajectory + Phase Optimization	Passive	No
[30]	Secure 6G Networks	RL	UAV Swarm Formation	Passive	Yes
[18]	Low-Altitude Economy	DRL	SWIPT + Security Optimization	Active	Yes
[27]	Distributed UAV Networks	Federated Learning	Trajectory + Phase Aggregation	Passive	Partial
[38]	Multi-UAV Systems	Learning-Based Optimization	Cooperative Resource Allocation	Passive	Yes
[22]	ISAC Systems	Transfer Learning + DRL	Hybrid-RIS ISAC Control	Hybrid	No
[33]	Maritime Communications	DRL	Energy Harvesting + Anti-Jamming	Passive	Yes
[40]	Secure Communications	RL	Anti-Jamming Optimization	Passive	No
[28]	mmWave Systems	MAB	Dual-Objective Phase Control	Passive	Yes
[36]	mmWave UAV-RIS	Energy-Aware MAB	Beam Selection	Passive	Yes
[34]	WiGig Networks	Contextual MAB	WiGig Enhancement	Passive	No
[35]	Aerial RIS Systems	Advanced MAB	Phase Selection	Passive	No
[26]	Aerial Active-RIS	Meta-RL	Resource Allocation	Active RIS	No

with time-varying cascaded channels, Doppler shifts, and fluctuating interference. Although RL and MAB frameworks can adapt to environmental changes, convergence speed and inference latency remain major concerns for real-time deployment. Future research should focus on ultra-fast adaptation through meta-learning that can facilitate rapid policy updates when channel conditions or user distributions change, significantly reducing retraining overhead. Developing lightweight online learning algorithms with bounded computational complexity is equally important to ensure real-time feasibility on resource-constrained aerial RIS platforms.

3) **Energy-Constrained Intelligence:** Aerial RIS platforms are severely constrained by energy and payload limits. Although energy-aware RL/MAB methods exist, unified frameworks that jointly optimize propulsion, communication, battery dynamics, and RIS control are still lacking. Future research must develop integrated trajectory–battery–reflection optimization models that explicitly address the energy–computation trade-off. Moreover, combining SWIPT and EH with lightweight AI requires multi-objective learning to balance throughput, harvested energy, and propulsion cost, supported by battery-aware policies, adaptive power allocation, and dynamic energy budgeting for sustainable UAV operation.

4) **Hardware-Aware and Edge-Assisted Learning:** Most existing studies assume ideal processing capabilities and abundant computational resources. However, UAV-mounted RIS controllers operate on embedded processors with limited memory, restricted computational throughput, and strict power budgets. Consequently, effective deployment demands hardware-aware AI design that explicitly accounts for processing latency, memory footprint, and energy consumption. Key research challenges include developing TinyML-based phase-control mechanisms suitable for microcontroller-level deployment, designing quantized and compressed DRL policies for efficient embedded inference, and developing adaptive edge–cloud collaboration strategies that intelligently partition computation between on-board processors and MEC infrastructure. Co-designing hard-

ware and learning algorithms is imperative to bridge the gap between simulation-driven research and real-world UAV-mounted RIS implementation, enabling scalable, reliable, and EE aerial intelligent surfaces for future 6G systems.

5) **Multi-UAV and Distributed Coordination:** Future 6G networks are expected to deploy multiple UAV-mounted RIS platforms operating cooperatively to extend coverage, enhance reliability, and improve spectral efficiency. However, distributed coordination among multiple aerial RIS units introduces substantial challenges, including synchronization, signaling overhead, limited backhaul capacity, and system stability under dynamic mobility conditions. The coupling between trajectory planning, RIS phase configuration, and inter-UAV interference further increases optimization complexity in multi-agent environments. Promising research directions include FL frameworks that operate under intermittent connectivity and limited bandwidth, Sparse GNN for topology-aware, scalable coordination, and distributed MAB algorithms for low-complexity resource allocation. Achieving scalable, resilient, and EE coordination among multiple aerial RIS platforms remains a fundamental open problem for practical large-scale deployment [37].

6) **Robustness and Security in Adversarial Environments:** UAV-mounted RIS systems are inherently vulnerable to security threats such as jamming, spoofing, eavesdropping, and adversarial manipulation, given their open wireless environment and aerial deployment. Although anti-jamming RL strategies have demonstrated promising performance, the robustness and reliability of lightweight AI models under adversarial conditions remain insufficiently explored. In particular, compressed or quantized models may be more susceptible to adversarial perturbations, raising critical concerns for safety-sensitive applications. Future research must prioritize developing secure, explainable, lightweight AI frameworks that maintain performance under intentional attacks. This includes adversarially robust RL, secure SWIPT and ISAC optimization mechanisms, and resilient distributed learning architectures.

7) **Standardization and Experimental Validation:** The gap between theoretical modeling and practical implementation poses a significant barrier to large-scale deployment. Real-world environments introduce non-idealities, including hardware nonlinearities, synchronization errors, channel estimation inaccuracies, and dynamic interference patterns, which are often overlooked in simulation-based evaluations. Future research must therefore prioritize prototype development of embedded RIS controllers suitable for UAV integration, along with real-time field trials in challenging environments such as maritime and NTN-integrated scenarios. The creation of standardized benchmark datasets tailored to lightweight UAV-RIS learning is also crucial for fair performance evaluation and reproducibility. In parallel, efforts to standardize aerial RIS architectures, control protocols, and interoperability frameworks will be necessary to ensure ecosystem compatibility. Bridging the gap between simulation and real-world experimentation is essential to translating lightweight AI-based UAV-mounted RIS frameworks into reliable, commercially viable 6G solutions.

8) **Integration with Orthogonal Time Frequency Space (OTFS) and High-Mobility Waveforms:** Recent studies have demonstrated the effectiveness of OTFS modulation in highly dynamic UAV communication scenarios, particularly for mobility-aware handover and high-speed communication systems [43], [44]. While OTFS-enabled UAV networks have been investigated, the integration of OTFS with UAV-mounted RIS systems remains largely unexplored. In UAV-mounted RIS architectures, cascaded BS-RIS-UAV and BS-RIS-user channels introduce additional delay-Doppler coupling, phase-alignment complexity, and reflection-induced channel sparsity that are absent in conventional UAV systems. The joint optimization of RIS phase configuration, UAV trajectory, and delay-Doppler-domain resource allocation under OTFS signaling, therefore, leads to a new class of tightly coupled, high-dimensional optimization problems. Addressing these challenges requires developing lightweight AI frameworks capable of delay-Doppler-aware BF, exploiting sparse channels, and enabling real-time trajectory adaptation under strict onboard energy and computational constraints.

V. CASE STUDY: MAB SCHEMES FOR UAV-MOUNTED RIS

Consider a UAV-mounted RIS that is deployed to assist a mmWave BS in serving multiple distributed hotspots. The UAV-mounted RIS dynamically moves between hotspots to enhance signal coverage for users experiencing non-line-of-sight conditions. Each hotspot has uncertain traffic demand and varying channel conditions, while the UAV operates under strict battery constraints. The objective is to design a trajectory that maximizes the cumulative achievable data rate across served hotspots while ensuring that the total propulsion and hovering energy does not exceed the UAV's battery capacity. The challenge lies in making sequential service decisions without full CSI and under limited onboard computational capability.

To address this, the trajectory design problem is reformulated as MAB framework, where each hotspot represents an arm and the UAV acts as the learner. At each decision step, the UAV

selects the next hotspot to visit based on observed rewards (achieved data rates) and contextual information, including hotspot location, historical payoffs, and residual energy. Classical Upper Confidence Bound (UCB) strategies prioritize arms with high uncertainty, while Thompson Sampling (TS) explores probabilistically via posterior sampling. Contextual TS (CTS) further enhances performance by incorporating side information into a Bayesian linear reward model, enabling the UAV to adapt its decisions to spatial and traffic features. Compared to DRL, these MAB schemes are computationally lightweight, require no extensive training datasets, and naturally balance exploration-exploitation under real-time constraints, making them suitable for energy-limited UAV-mounted RIS platforms.

Simulation results shown in Fig.2 demonstrate the practical importance of MAB-based lightweight AI. The CTS scheme achieves data rate [Gbps] performance exceeding 90% of the optimal benchmark while significantly outperforming classical UCB and TS, nearest-hotspot selection, and random policies. In terms of EE [Gbps/mJ], defined as the division of the data rate over the consumed energy per hotspot, CTS shows orders-of-magnitude improvements over non-contextual or greedy strategies, especially as the number of hotspots increases or the covered region expands. These results confirm that MAB-based decision-making can effectively extend UAV operational lifetime while preserving high communication throughput. Therefore, MAB schemes provide a scalable, low-complexity, and energy-aware intelligence layer for real-time trajectory optimization in airborne RIS systems.

VI. CONCLUSION

This paper provides an overview of lightweight AI techniques for UAV-mounted RIS systems and their role in enabling scalable, EE, and real-time optimization for 6G networks. UAV-mounted RIS enhances coverage, spectral efficiency, and support for emerging applications, including ISAC, SWIPT, NTN integration, and high-mobility communications. However, stringent energy, memory, and computational constraints on UAV platforms necessitate the development of efficient learning frameworks. Hence, we examined lightweight AI techniques, model-efficiency techniques (pruning, quantization, distillation, TinyML), adaptive decision-making methods (RL, meta-learning, MAB), topology-aware distributed learning (FD, sparse GNNs), and hybrid model-driven approaches (deep unfolding) in aerial RIS scenarios. Furthermore, we presented a case study that highlights the importance of the MAB scheme as a lightweight AI for optimizing the trajectory of UAV-mounted RISs with EE considerations. Future work will prioritize hardware-software co-design, embedded implementation, and experimental validation. Integrating UAV-mounted RIS with advanced 6G technologies such as OTFS and distributed edge intelligence will require structured, energy-aware, and delay-sensitive learning architectures.

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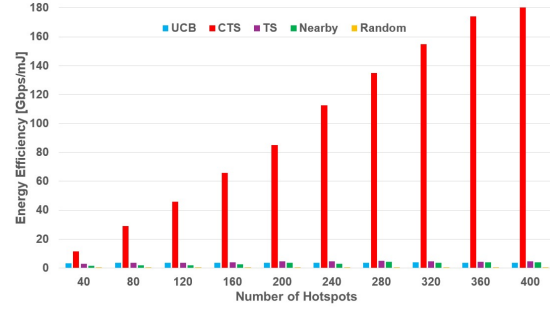
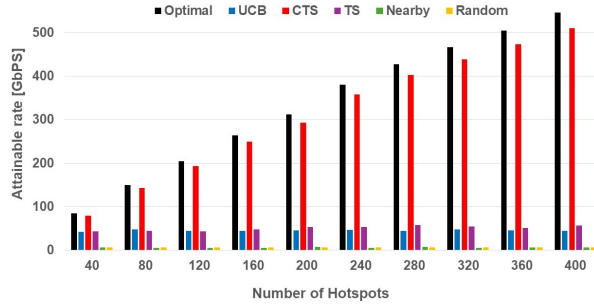


Fig. 2. Attainable rate and EE Comparison of CTS against UCB, TS, nearby, and Random schemes at various hotspot numbers and 128 RIS elements within $4 \times 4 \text{ Km}^2$ and UAV height of 50m.

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